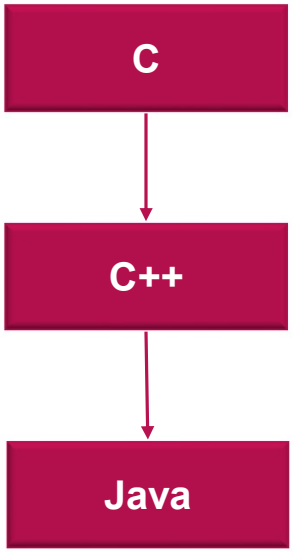
Objectives

- ✓ Evolution of Java.
- ✓ Java Virtual Machine(JVM).
- ✓ Java Execution Engine

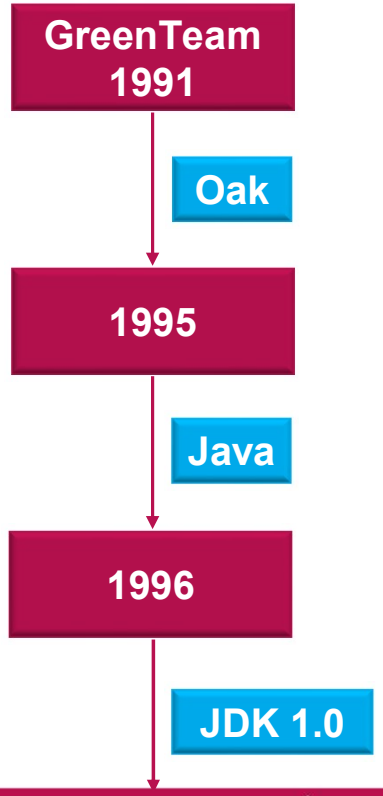
Evolution of Java

- ✓ Java is related to the two programming languages
  - ✓ C
  - ✓ C++
- ✓ Java takes syntax from C.
- ✓ Object oriented principles influenced from C++.



Java Creation

- ✓ In 1991 the Green Team of Sun Microsystem lead by **James Gosling** developed Java Programming language.
- ✓ Initially language was known as Oak.
- ✓ In 1995 renamed as Java.
- ✓ JDK 1.0 released in 1996.
- ✓ Sun microsystem is now a subsidiary of Oracle Corporation.



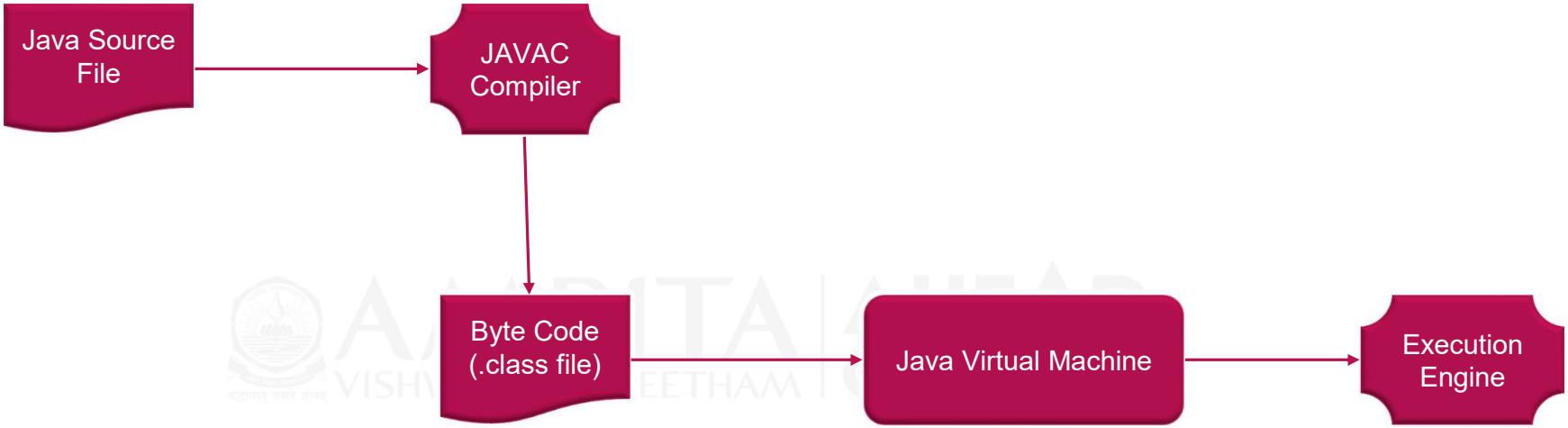
Java Bytecode

- ✓ Java bytecode is the output of Java compilation.
- ✓ Bytecode is executed by Java runtime system.
- ✓ All Java platforms consist of a Java Virtual Machine (JVM) and an Application Programming interface (API).

Java Virtual Machine

- ✓ The Java Virtual Machine is a program, for a particular hardware and software platform, that runs Java technology applications.
- ✓ An API is a collection of software components that you can use to create other software components or applications.
- ✓ JVM makes Java programs portable and secure.

Java Virtual Machine



Java Execution Engine

- ✓ Java execution engine executes the bytecode by reading the bytecode in units, like a machine reading code lines one by one.

- ✓ The Execution Engine has three major components
  - ✓ Interpreter - executes the bytecode in a sequential method
  - ✓ JIT Compiler - enhances the Java Application performance at run time
  - ✓ Garbage Collector – JVM heap space will be searched for the possible objects in use and removes the unused ones.

Takeaways

- ✓ Java evolution.
- ✓ Java Bytecode and Java Virtual Machine.
- ✓ Java Execution Engine.

Reference

[1] Herbert Schildt, *"Java: The Complete Reference, Eleventh Edition", Oracle 2018.*